
ASSIGMENT

PROGRAMMING USING C

WEEK – 05

NAME : K.Venkateshwari
ROLL NO : 2116240801373
DEPT & SEC :ECE F

Week-5(a)

Write a program that prints a simple chessboard.

```
#include<stdio.h>
int main()
{
    int t,d,i=0,il,iz,o;
    char c;
    scanf("%d",&t);
    while(i<t)
    {
        scanf("%d",&d);
        il=0;
        while(il<d)
        {
            o=1;
            iz=0;
            if(il%2==0)
            {
                o=0;
            }
            while(iz<d)
            {
                c='B';
                if(iz%2==o)
                {
                    c='W';
                }
                printf("%c",c);
                iz++;
            }
            il+=1;
            printf("\n");
        }
        i++;
    }
}
```

	Input	Expected	Got	
✓	2 3 5	WBW BWB WBW WBWBW BWBWB WBWBW BWBWB WBWBW	WBW BWB WBW WBWBW BWBWB WBWBW BWBWB WBWBW	✓

Passed all tests! ✓

Write a program that takes input:

The first line contains T, the number of test cases

Each test case contains an integer N and also the starting character of the chessboard

```
#include<stdio.h>
int main()
{
    int t,d,i,il,iz,o,z;
    char c,s;
    scanf("%d",&t);
    for(i=0;i<t;i++)
    {
        scanf("%d %c",&d,&s);
        for(il=0;il<d;il++)
        {
            z=(s=='W')?0:1;
            o=(il%2==z)?0:1;
            for(iz=0;iz<d;iz++)
            {
                c=(iz%2==o)?'W':'B';
                printf("%c",c);
            }
            printf("\n");
        }
    }
}
```

	Input	Expected	Got	
✓	2 2 W 3 B	WB BW BWB WBW BWB	WB BW BWB WBW BWB	✓

Passed all tests! ✓

Decode the logic and print the Pattern that corresponds to given input.

If N = 3

then pattern will be :

```
10203010011012
**4050809
****607
```

If N = 4, then pattern will be:

```
1020304017018019020
**50607014015016
****809012013
*****10011
```

```
#include<stdio.h>
int main()
{
    int n,v,p,c,i,in,il,iz,tl,t;
    scanf("%d",&t);
    for(tl=0;tl<t;tl++)
    {
        v=0;
        scanf("%d",&n);
        printf("Case #%d\n",tl+1);
        for(i=0;i<n;i++)
        {
            c=0;
            if(i>0)
            {
                for(il=0;il<i;il++)
                {
                    printf("***");
                }
            }
            for(il=i;il<n;il++)
            {
                if(i>0)
                {
                    c++;
                }
                printf("%d0",++v);
            }
            if(i==0)
            {
                p=v+(v*(v-1))+1;
                in=p;
            }

            in=in-c;
            p=in;
            for(iz=i;iz<n;iz++)
            {
                printf("%d",p++);
                if(iz!=n-1)
                {
                    printf("0");
                }
            }
            printf("\n");
        }
    }
}
```

	Input	Expected	Got	
✓	3 3 4 5	Case #1 10203010011012 **4050809 ****607 Case #2 1020304017018019020 **50607014015016 ****809012013 *****10011 Case #3 102030405026027028029030 **6070809022023024025 ***10011012019020021 *****13014017018 *****15016	Case #1 10203010011012 **4050809 ****607 Case #2 1020304017018019020 **50607014015016 ****809012013 *****10011 Case #3 102030405026027028029030 **6070809022023024025 ***10011012019020021 *****13014017018 *****15016	✓

Passed all tests! ✓

Week-5(b)

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

```
#include<stdio.h>
#include<math.h>
int main()
{
    int n;
    scanf("%d",&n);
    int x=0,n2=n;
    while(n2!=0)
    {
        x++;
        n2=n2/10;
    }
    int sum=0;
    int n3=n,n4;
    while(n3!=0)
    {
        n4=n3%10;
        sum=sum+pow(n4,x);
        n3=n3/10;
    }
    if(n==sum)
    {
        printf("true");
    }
    else
    {
        printf("false");
    }
    return 0;
}
```

	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Take a number, reverse it and add it to the original number until the obtained number is a palindrome. Constraints $1 \leq \text{num} \leq 999999999$ Sample Input 1 32 Sample Output 1 55 Sample Input 2 789 Sample Output 2 66066

```
#include<stdio.h>
int main()
{
    int n,rn,nt=0,i=0;
    scanf("%d",&n);
    do
    {
        nt=n;
        rn=0;
        while(n!=0)
        {
            rn=rn*10+n%10;
            n=n/10;
        }
        n=nt+rn;
        i++;
    }
    while(rn!=nt || i==1);
    printf("%d",rn);
    return 0;
}
```

	Input	Expected	Got	
✓	32	55	55	✓
✓	789	66066	66066	✓

Passed all tests! ✓

A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

```
#include<stdio.h>
int main()
{
    int n=1,i=0,nt,co=0,e;
    scanf("%d",&e);
    while(i<e)
    {
        nt=n;
        while(nt!=0){
            co=0;
            if(nt%10!=3&&nt%10!=4)
            {
                co=1;
                break;
            }
            nt=nt/10;
        }
        if(co==0)
        {
            i++;
        }
        n++;
    }
    printf("%d",--n);
    return 0;
}
```

	Input	Expected	Got	
✓	34	33344	33344	✓

Passed all tests! ✓